

Submodular Maximization over Many Matroids via Ordered Local Search

Neta Singer

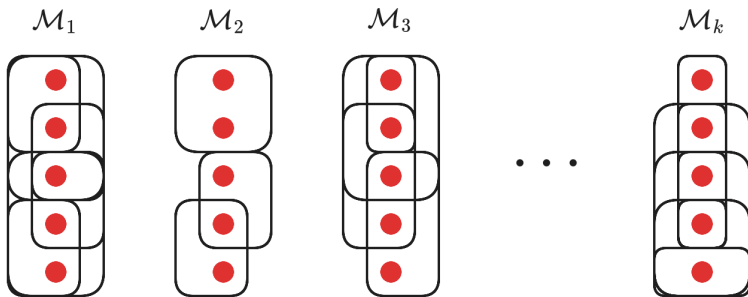
École Polytechnique Fédérale de Lausanne

Joint work with Theophile Thiery

September 22, 2026

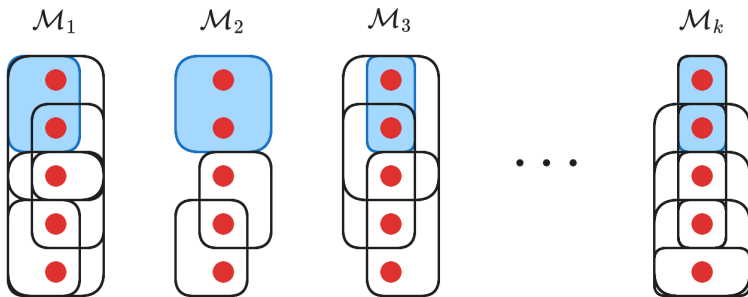
The problem:

- **Input:** k matroids over E , submodular function $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$

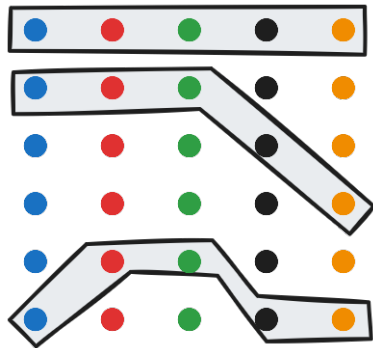
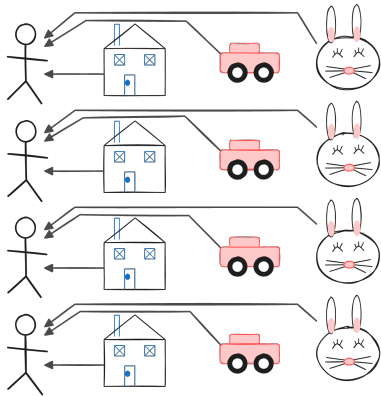


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- **Input:** k matroids over E , submodular function $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$
- **Goal:** find set $S \subseteq E$ in all k matroids with $f(S)$ maximized



The motivation:



Prior guarantees:

- Approximation algorithm where output A satisfies $w(OPT) \leq \alpha \cdot w(A)$.
- Hardness of approximation by a factor $k/12$ [LST'24].

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Our results:

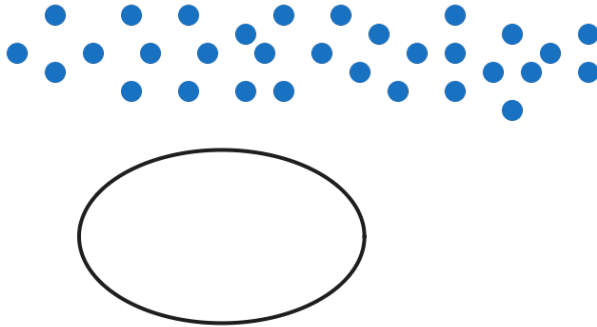
Theorem (S., Thiery 2026+)

*There exists a $\frac{k}{2} + o(k)$ -approximation algorithm for **monotone submodular maximization** over a k -Matroid Intersection.*

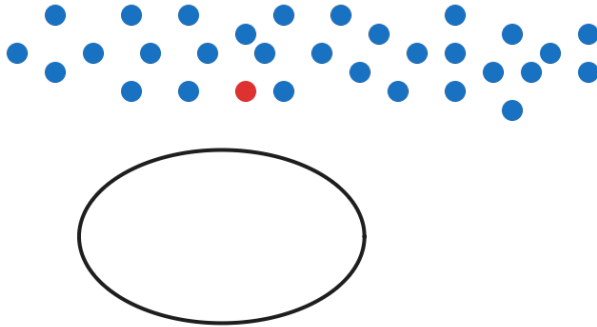
Theorem (S., Thiery 2026+)

*There exists a $\frac{k}{2.16} + o(k)$ -approximation algorithm for **weighted k -Set Packing**.*

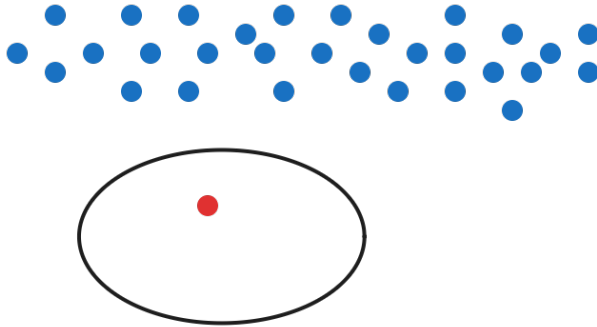
A typical local search:



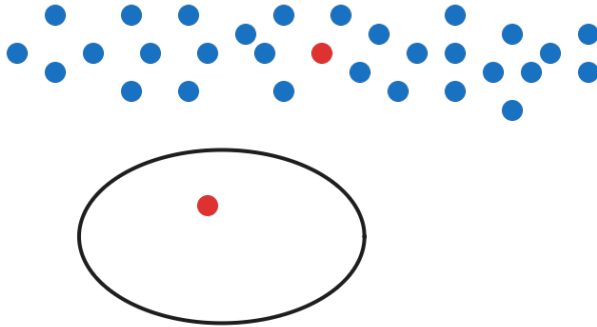
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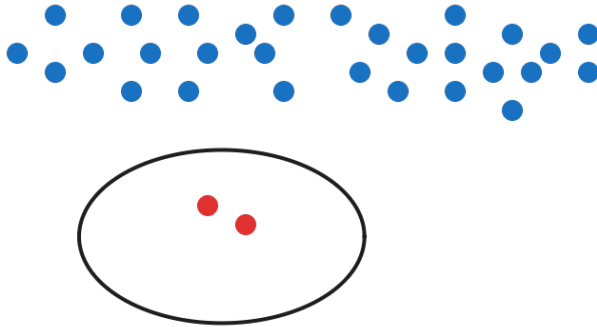
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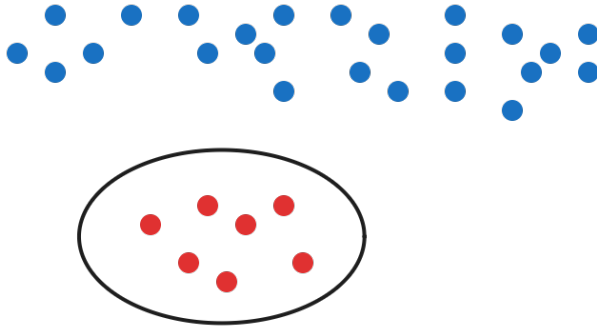
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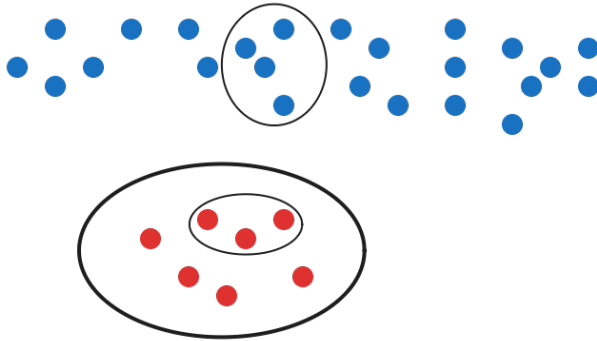
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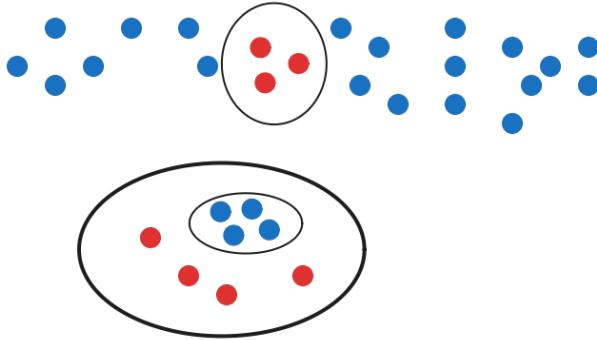
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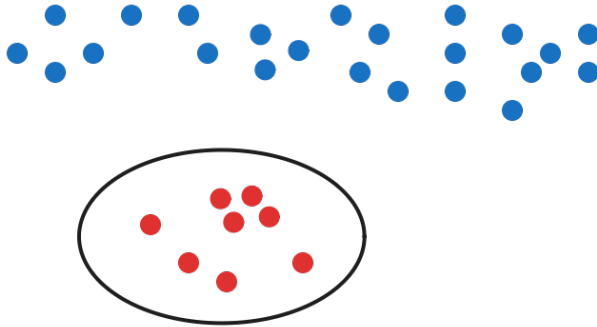
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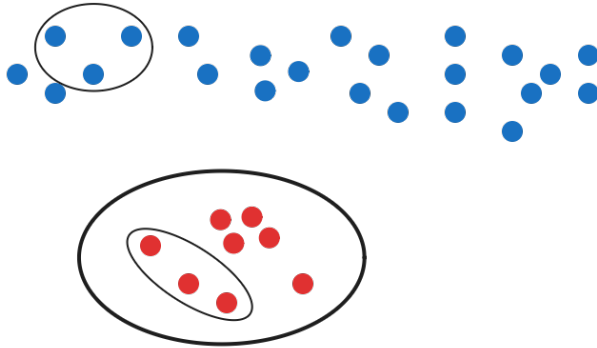
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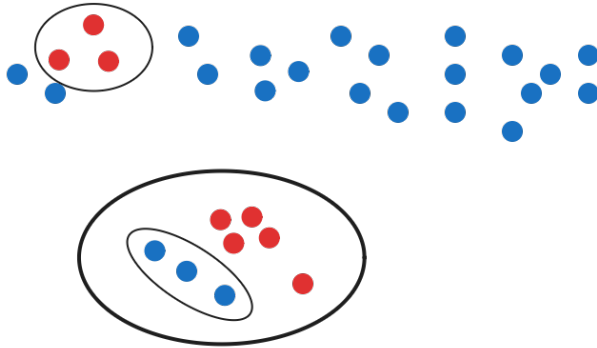
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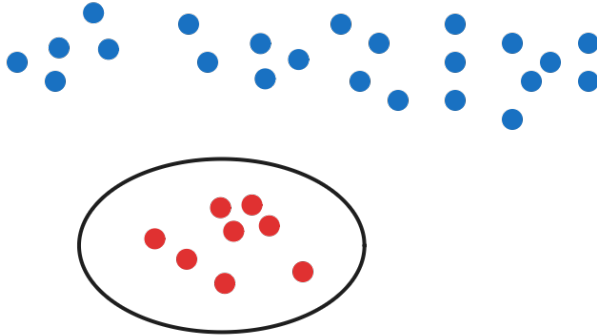
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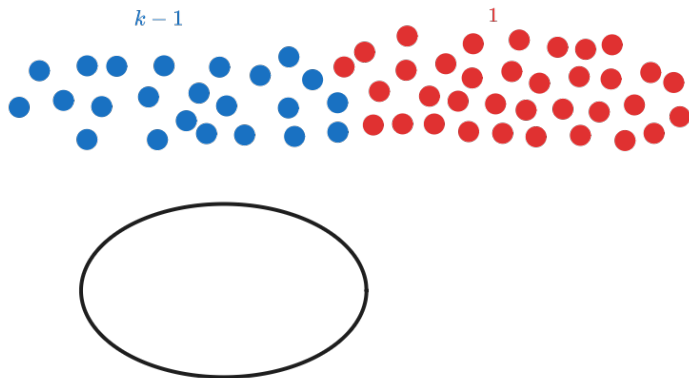
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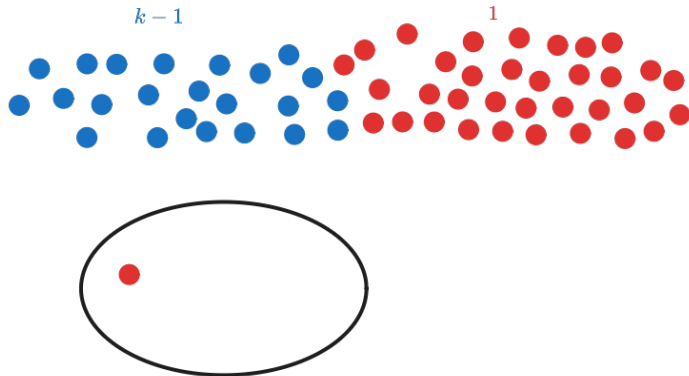
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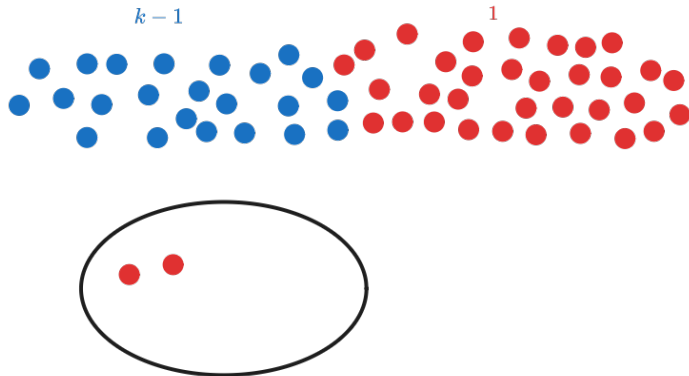
An illustrative bad example:



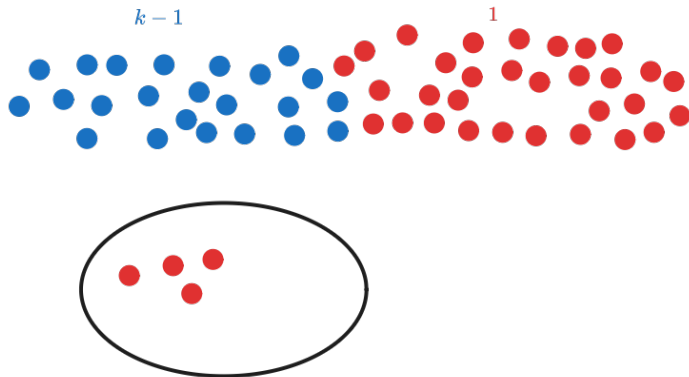
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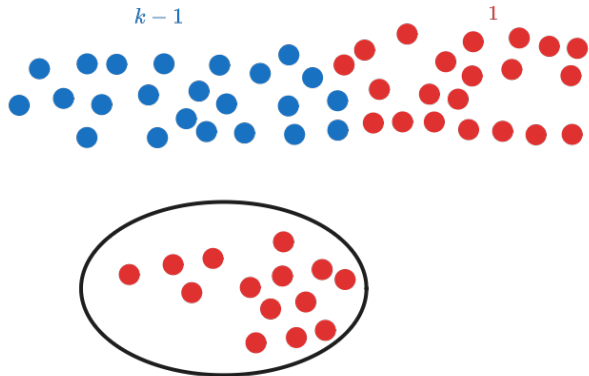
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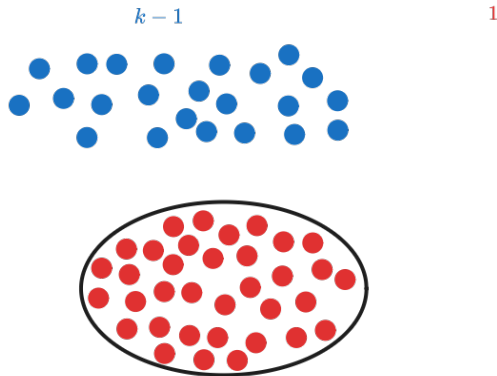
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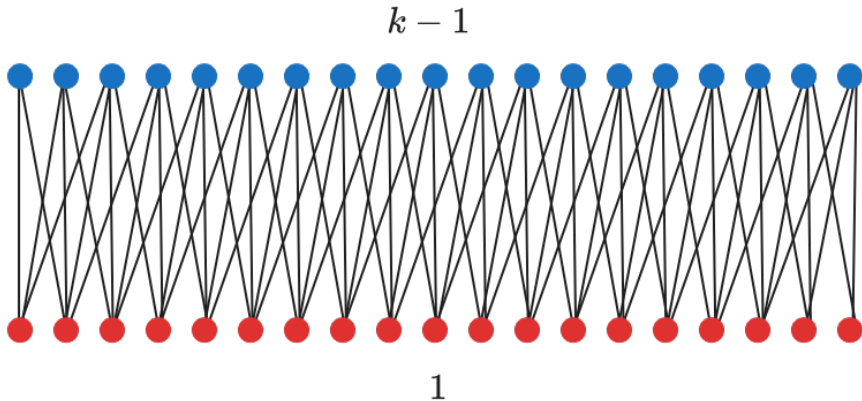
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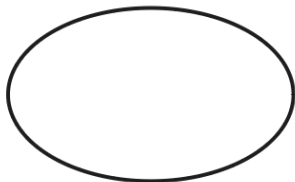
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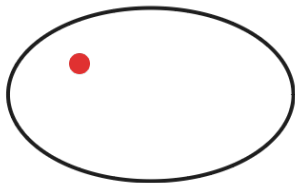
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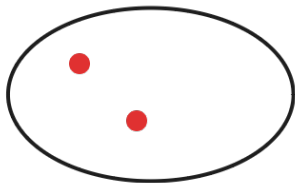
Takeaway 1: order matters!



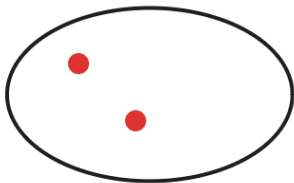
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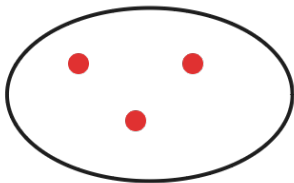
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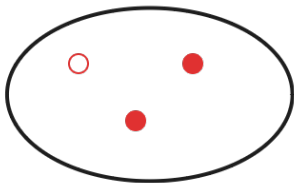
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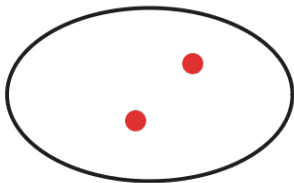
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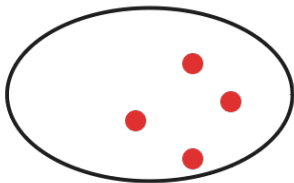
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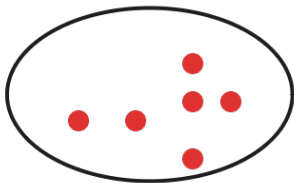
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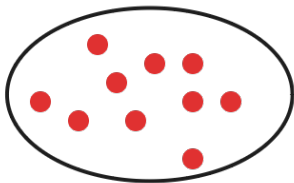
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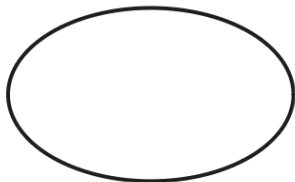
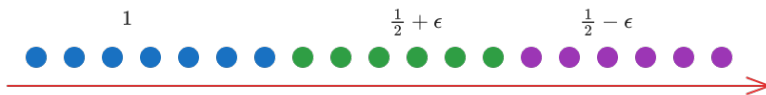
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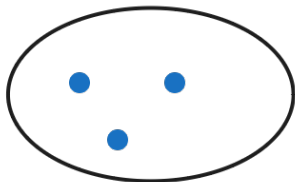
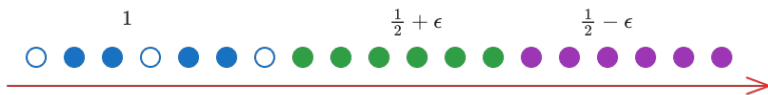
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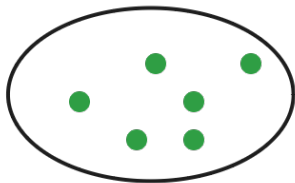
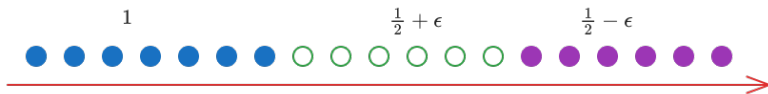
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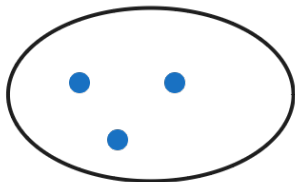
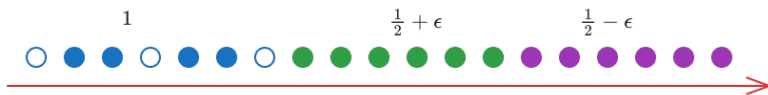
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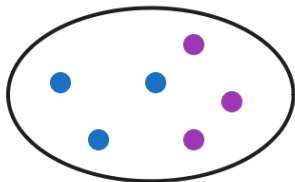
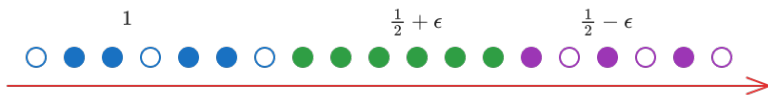
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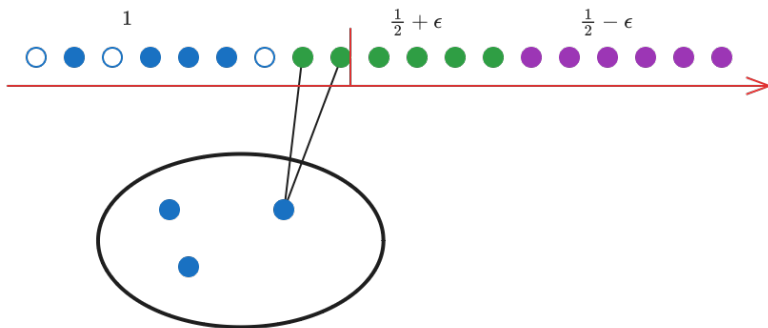
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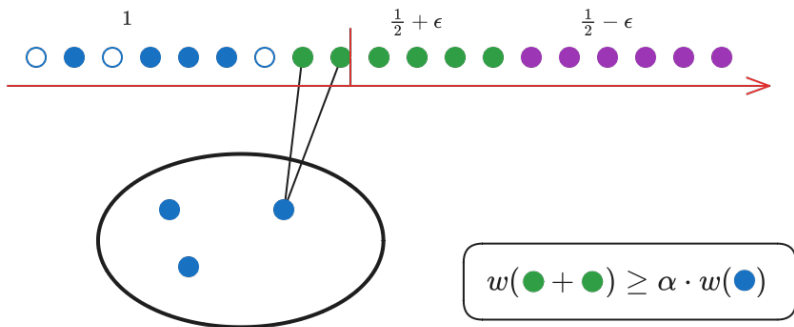
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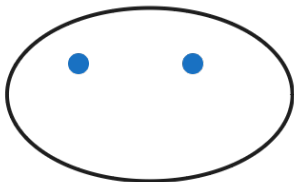
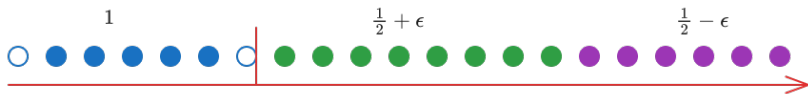
Takeaway 2: large gains are needed per swap



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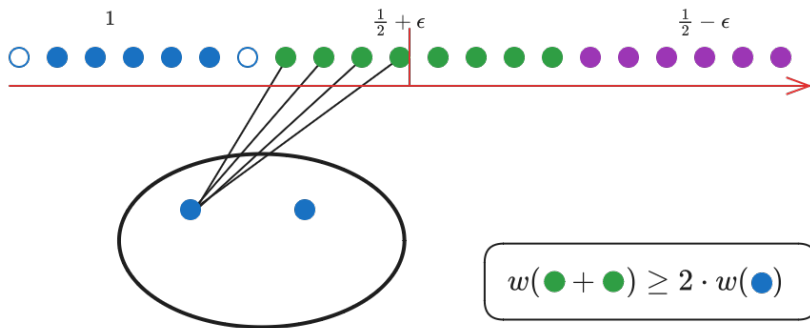


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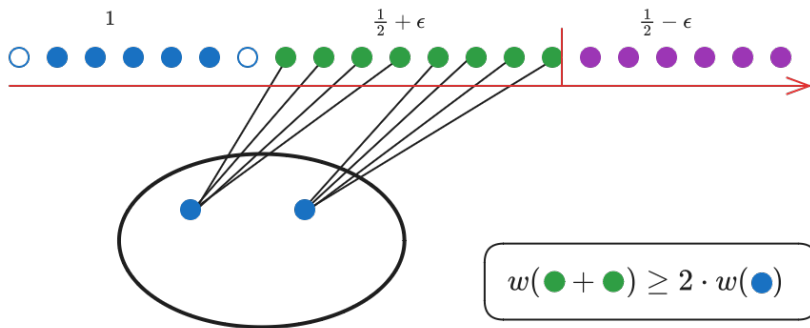


$$w(\bullet + \bullet) \geq 2 \cdot w(\bullet)$$

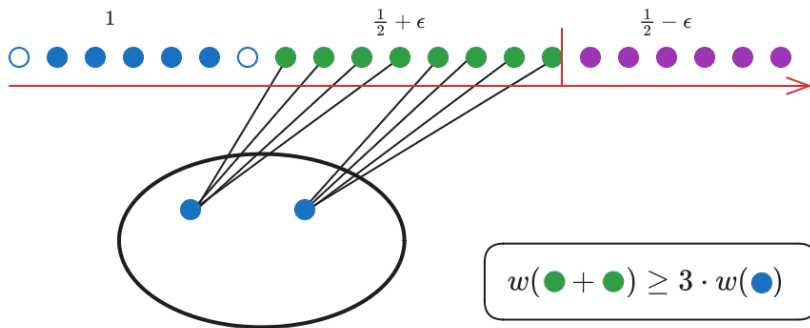
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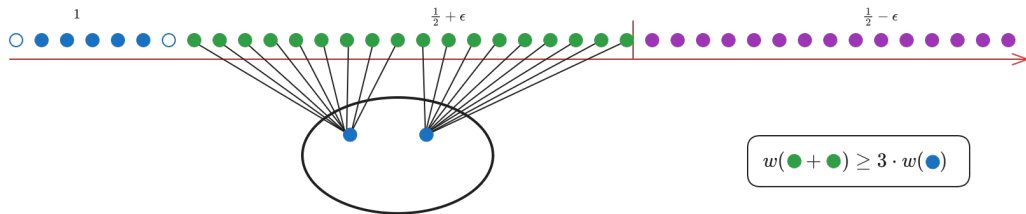
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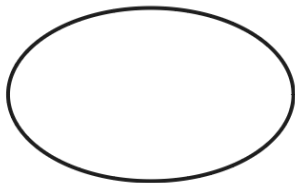
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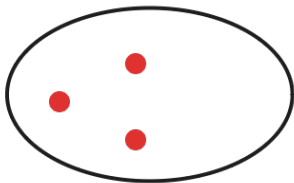
An illustrative bad example:



The full algorithm with $\alpha = \sqrt{k \log k}$:

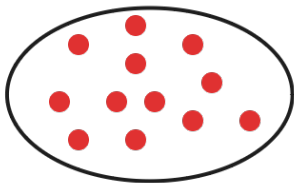


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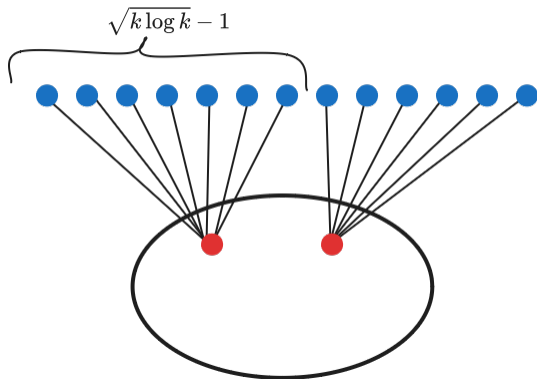
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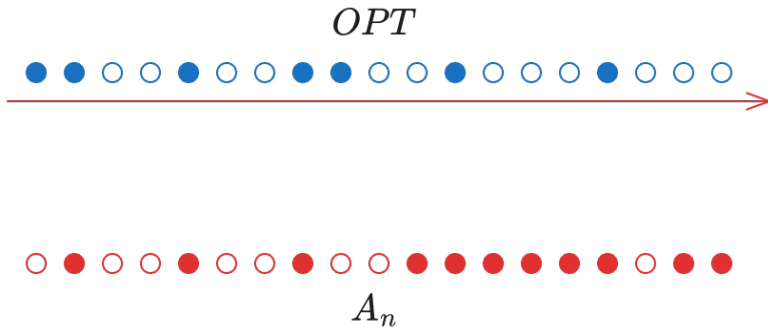


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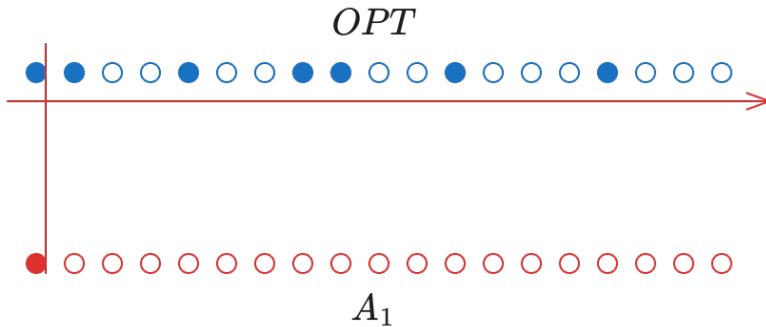
Takeaway 3: strategy is right when $k \rightarrow \infty$



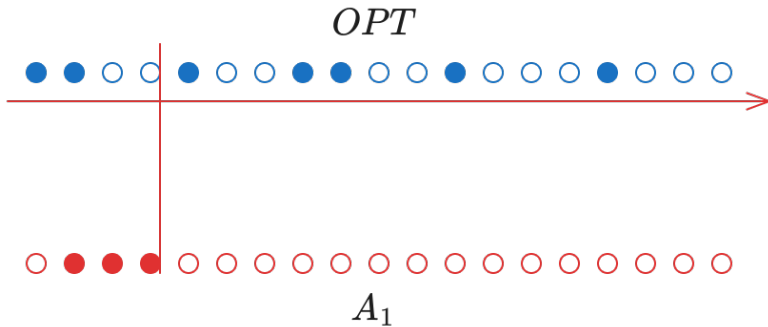
A quick breakdown of the analysis:



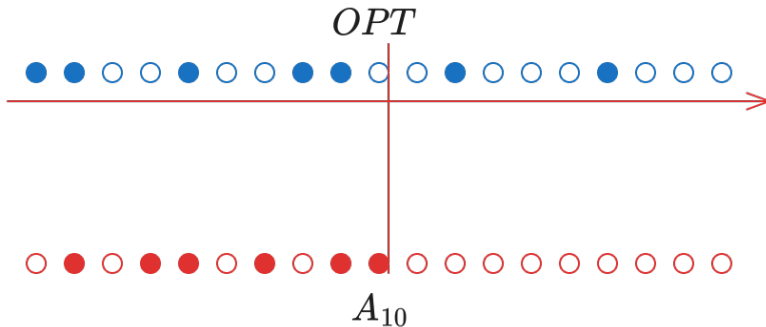
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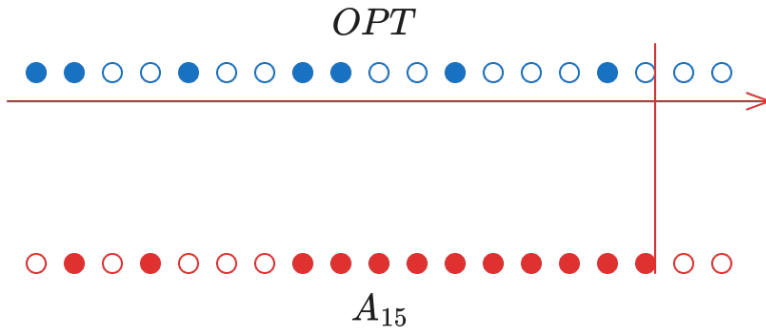
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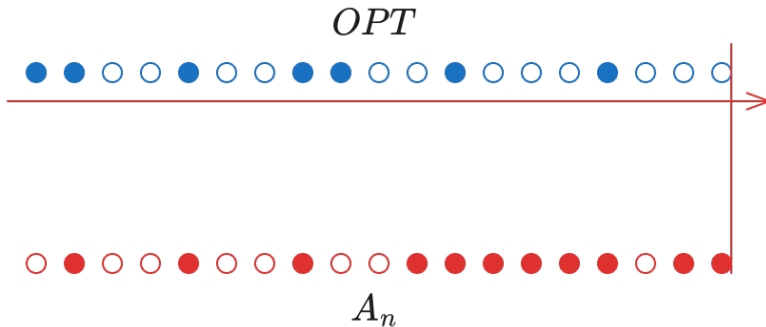
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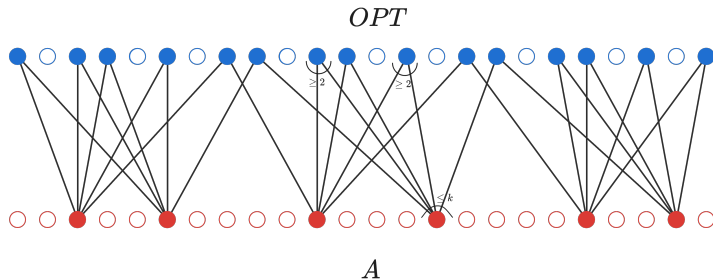
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OPT

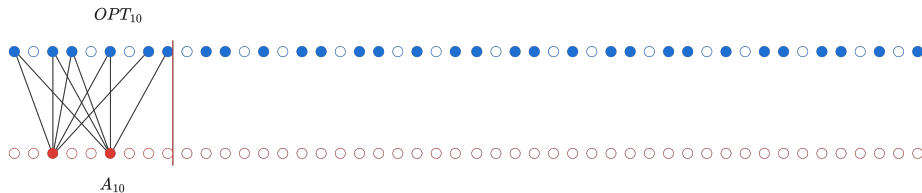


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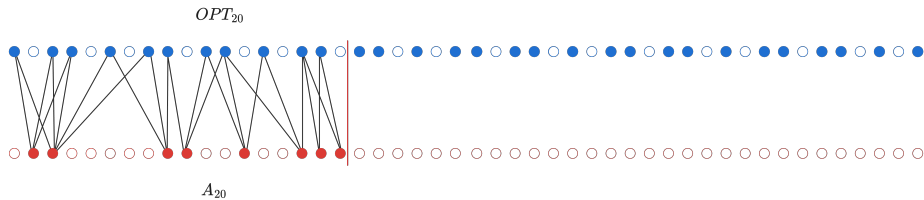
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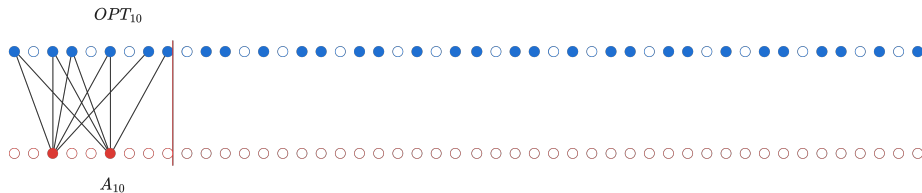
A quick breakdown of the analysis:

Lemma (S., Thiery 2026+)

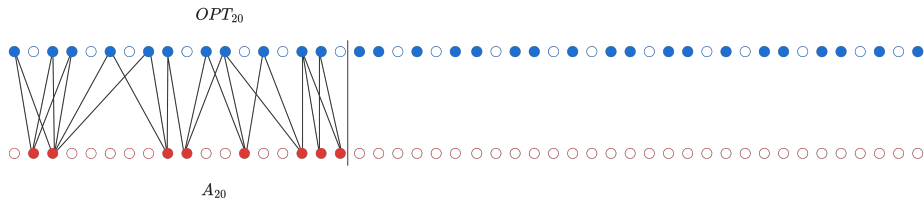
For **every** τ , we have $OPT_\tau = OPT_\tau^{(relevant)} + OPT_\tau^{(irrelevant)}$ and

$$\left| OPT_\tau^{(relevant)} \right| \leq \frac{k}{2} |A_\tau| \quad \text{and} \quad \left| OPT_\tau^{(irrelevant)} \right| \leq \sum_{a \in A_\tau} \alpha \frac{w(a)}{\tau}.$$

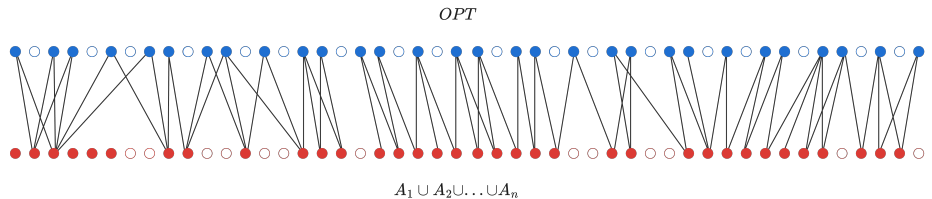
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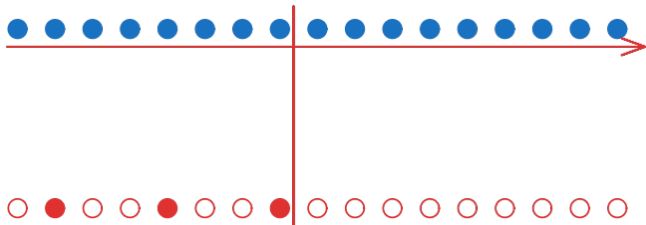
A quick breakdown of the analysis:

- **Problem:** get $k/2$ apx with respect to $A_1 \cup \dots \cup A_n$ instead of A_n .
- **Observation:** every item removal corresponds to adding items of value $\alpha = \sqrt{k \log k}$ factor more.

$$w(A^{t+1}) - w(A^t) \geq (\alpha - 1)w(R_t) \implies w(\text{removed elements}) \leq \frac{w(A_n)}{\alpha - 1}$$

Dealing with submodularity:

- $f_i(S) = f(S|X_i - S)$
- X_i = all items used in a solution by time i



Future directions:

Theorem (S., Thiery 2026+)

*There exists a $\frac{k}{2} + o(k)$ -approximation algorithm for **monotone submodular maximization** over a k -Matroid Intersection.*

- Small k regime?
- Better run times?
- General independence systems?
- Other constraints?
- New local search “frameworks”?

Thank you!

References



Jon Lee, Maxim Sviridenko, and Jan Vondrák (2010)

Matroid matching: the power of local search.

STOC pages 369-378.



Justin Ward (2012)

A $(k+3)/2$ -approximation algorithm for monotone submodular k -set packing and general k -exchange systems.

STACS.



Jon Lee, Maxim Sviridenko, and Jan Vondrák (2009)

Submodular maximization over multiple matroids via generalized exchange properties.

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Meike Neuwöhner (2023)

Passing the limits of pure local search for weighted k -set packing.

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Marek Cygan (2013)

Improved approximation for 3-dimensional matching via bounded path- width local search.

Foundations of Computer Science, FOCS 2013 pages 509-518.



Euiwoong Lee, Ola Svensson, Theophile Thiery (2024)

Asymptotically Optimal Hardness for ℓ_1 -Set Packing and ℓ_1 -Matroid Intersection.



Neta Singer, Theophile Thiery (2025)

Better Approximation for Weighted k Matroid Intersection.



Neta Singer, Theophile Thiery (2026)

Submodular Maximization over Many Matroids via Ordered Local Search.



Moran Feldman, Justin Ward (2026)

Submodular Maximization over a Matroid k -Intersection: Multiplicative Improvement over Greedy.